Quantifying the Privacy Implications of High-Fidelity Synthetic Network Traffic

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Abstract

To address the scarcity and privacy concerns of network traffic data, various generative models have been developed to produce synthetic traffic. However, synthetic traffic is not inherently privacy-preserving, and the extent to which it leaks sensitive information, and how to measure such leakage, remain largely unexplored. This challenge is further compounded by the diversity of model architectures, which shape how traffic is represented and synthesized. We introduce a comprehensive set of privacy metrics for synthetic network traffic, combining standard approaches like membership inference attacks (MIA) and data extraction attacks with network-specific identifiers and attributes. Using these metrics, we systematically evaluate the vulnerability of different representative generative models and examine the factors that influence attack success. Our results reveal substantial variability in privacy risks across models and datasets. MIA success ranges from 0% to 88%, and up to 100% of network identifiers can be recovered from generated traffic, highlighting serious privacy vulnerabilities. We further identify key factors that significantly affect attack outcomes, including dataset diversity and how well the generative model fits the training data. These findings provide actionable guidance for designing and deploying generative models that minimize privacy leakage, establishing a foundation for safer synthetic network traffic generation. We release our code and evaluation framework to support reproducibility and facilitate future research.

CCS Concepts

• **Security and privacy → Network security**.



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1 Introduction

Network traffic generators have gained significant traction in both academia and industry due to their versatility [1–4]. They can augment training data for underrepresented scenarios (such as network attacks), enabling more robust anomaly-detection models [5, 6]. They also provide a practical way to generate diverse, realistic traffic for analysis, algorithm testing, and performance benchmarking—allowing users to work with high-fidelity data without needing access to raw traffic traces [1, 5, 7].

However, generative modeling does not automatically come with privacy [8–10]. While prior work has focused largely on fidelity [3, 4, 11], i.e., how realistic the generated traffic appears, much less attention has been paid to the privacy implications of synthetic traffic generation. In practice, many generative models either overlook privacy risks entirely [11, 12] or offer only minimal protection against them. Compounding this issue, systematic evaluation of privacy risks in synthetic network traffic remains scarce. A few existing efforts primarily rely on existing membership inference attacks (MIA) [3, 13, 14] as the sole measure of privacy leakage, even though network traffic contains highly sensitive information beyond data membership [1, 15–17].

To bridge the gap between real-world privacy risks in network traffic and the limited metrics currently used to evaluate them, we introduce a comprehensive suite of privacy metrics spanning three complementary attack types: (1) membership inference attacks (MIA), (2) data extraction attacks, and (3) network-specific privacy attacks. MIA and data extraction are standard privacy attacks against generative

models [18–20]. MIA determines whether a traffic sample was included in the model's training data, while data extraction measures whether the model memorizes and reproduces training data. However, neither attack reveals what information is exposed, whether it is sensitive, or the extent of the leakage, limiting their interpretability for network privacy.

Leveraging insights from real-world privacy attacks on network traffic, including fingerprinting [21, 22], attribute inference [17, 23], and topology reconstruction [24, 25], we introduce network-specific metrics that capture three key risks: (i) network identifier memorization (e.g., IP and MAC addresses), which can directly expose users and devices; (ii) sensitive property leakage, which can reveal user behaviors, visited websites, and application usage patterns; and (iii) topology extraction, which exposes structural relationships between network entities and can compromise organizational security. Together, these metrics capture the practical privacy risks of synthetic traffic generation and inform deployment decisions.

We evaluate our metrics across four state-of-the-art generative models, each representing a distinct generation paradigm (GAN, diffusion, transformer, SSM), and five datasets collected under diverse network conditions and tasks. Here is the summary of our research questions:

- **RQ1: How successful can each privacy attack be under the most challenging conditions for an attacker? (§5)** We evaluate MIA, data extraction, and network-specific privacy attacks across network generative models under *the most challenging conditions for an attacker: attackers only have black-box access and each generative model only undergoes minimal training* (i.e., less likely to overfit to or memorize training data). We find that even under these constrained conditions, MIA attacks, particularly against NetSSM and NetShare, can achieve true positive rate (TPR) exceeding 0.8 on some datasets. Additionally, up to 100% of network identifiers in training data can be correctly recovered from the outputs of NetSSM and TrafficLLM.
- **RQ2: Which factors impact the success of privacy attacks? (§6)** We evaluate the factors that could influence each privacy attack type. These include attacker capabilities (black-box vs. white-box access), model overfitting, and data properties (training data diversity). Our findings reveal that for MIA, white-box access can increase the TPR by nearly 0.2 on average for NetDiffusion and NetSSM. For data extraction and network-specific attacks, models with greater overfitting increase extractable rate and network identifiers' precision and recall scores.
- **RQ3: What is the privacy-utility tradeoff when applying privacy risks mitigation techniques? (§6)** We explore whether mitigation strategies such as identifier anonymization and noise addition can reduce privacy risks, and examine the privacy-utility tradeoffs of these approaches. Our findings show that no single mitigation strategy can address all privacy risks and applying these techniques can reduce data utility.

The remainder of this paper is organized as follows. Section 2 covers related work. Section 3 describes privacy attack and mitigation methodology. Section 4 presents experimental setup. Section 5, 6, and 7 each covers results for a research question. Section 8 discusses implications of our findings and we conclude our work in Section 9.

2 Related Work

2.1 Network Traffic Generation

Synthetic trace generation has long been an important topic in network research, and consequently, a substantial body of work has been dedicated to advancing this area.

Non-generative models: Early network traffic generation relied on simulation tools [26–31] or statistical methods [32, 33]. However, simulation tools fail to capture the complexity of real-world traffic, while statistical methods generate only limited network features and lack the fine-grained granularity required for many network analysis tasks.

GAN-based models: Generative Adversarial Networks (GANs) are a class of generative models that train a generator and a discriminator in an adversarial manner until generated outputs become statistically indistinguishable from real data. GAN-based traffic generators include NetShare [3], DoppelGANger [34] and PacketCGAN [35]. NetShare first merges traffic collected from different measurement epochs into a single large trace and then splits the trace into flows based on five-tuples. Next, each header field is encoded using dataset-dependent encoding schemes. A time-series GAN is then trained on the encoded data to capture temporal dependencies within each flow. Generated outputs include flow-level statistics and selected packet header fields, including IP addresses, ports, TTL values, packet sizes, and inter-arrival times.

Diffusion-based models: Diffusion models are a class of generative AI models that generate high-quality data by learning to reverse a gradual noise-addition process. Diffusion-based network traffic generators [4, 36–38] often represent network traffic as images and fine-tune diffusion models to synthesize realistic traffic samples. NetDiffusion [4], for example, represents packet headers as 2D images by first converting header fields into bitmaps and then applying a three-value color encoding scheme to represent bit values of 0, 1, and missing or undefined fields. The resulting images are used to fine-tune a pre-trained Stable Diffusion model conditioned on traffic class labels. During generation, the model conditions on both a text prompt and a structural edge map extracted using ControlNet to better preserve structural

Model	Base Model	Granularity	Unit	Generated Data
NetDiffusion	Diffusion	Packet	Bit	Network layer (except IP addresses), transport layer for the first 1024 packets.
NetSSM	State-space model	Packet	Byte	Ethernet layer, network layer and transport layer for the entire PCAP file.
TrafficLLM	Transformer	Packet	Hex digits	Ethernet layer, network layer, and transport layer for the first packet.
NetShare	GANs	Flow	Tabular values	5-tuples and header fields including packet length, type of service, identification, control flag, data offset, time to live, and inter-arrival time.

Table 1: Overview of network traffic generative models and their output representations.

patterns, followed by a post-generation correcting process to ensure generated traffic follow network protocols.

Transformer-based models: Transformer-based models are neural networks that process sequential data by using a mechanism called self-attention, which allows the model to look at all elements in a sequence at once and learn how they relate to each other. Transformer-based traffic generators [12, 39, 40] typically tokenize network traffic into hexadecimal values or byte-level representations and then train or fine-tune transformer models to learn traffic patterns and representations. TrafficLLM [12] fine-tunes GPT-2 on tokenized network traffic using a two-stage pipeline. The first stage improves the model's instruction-following capability, while the second stage focuses on learning network traffic patterns and representations.

State-space models: State-space models (SSMs) [11] are mathematical frameworks for modeling sequential dynamic systems through hidden states that evolve over time. NetSSM [11] applies this framework to network traffic generation through a structured tokenization and pretraining pipeline. First, it tokenizes packet headers at the byte level, mapping each byte to an integer in the range [0, 255]. These token sequences are further augmented with control tokens, such as label tokens to indicate traffic type and boundary tokens to delimit individual packets. The resulting sequences are used to pretrain an SSM architecture that predicts the next token autoregressively given the preceding context.

In this study, we examine privacy leakage across four models - NetShare, NetDiffusion, TrafficLLM, and NetSSM — selected for two reasons. First, each model represents the state of the art within its respective generative paradigm: GANs, diffusion models, transformers, and state-space models. Second, this diversity of paradigms enables a comprehensive assessment of privacy risks across fundamentally different architectural approaches, facilitating a rigorous evaluation of the privacy-utility trade-offs inherent to each approach. Table 1 summarizes key differences between each model.

2.2 Privacy Risks of Generative Models

Membership inference attacks: Under MIA, an adversary can leverage generative models' tendency to overfit to their training data to differentiate between training versus non-training samples based on how the model behaves on them. In a typical MIA setting, attackers first extract observable signals and then apply a threshold or classifier to infer membership. In a white-box MIA where an attacker has access to the model's internal parameters and gradients, an attacker can use a wider variety of signals such as model loss [41, 42] and gradient on input samples [8, 43]. In contrast, in a black-box setting, membership signals observed by an attacker are often limited to the model's outputs and output token probabilities [44–46]. To overcome this limitation, an attacker can train shadow models on some shadow train set [46, 47], enabling them to learn membership inference patterns without direct access to the model's internals. MIA can be successful on a variety of models, including diffusion models [18, 43, 48, 49], transformer-based models [50–54] and GAN-based models [8, 46].

Data extraction attacks: Extraction attacks concern with whether an attacker can reconstruct training data, be it text sequences [44, 55] or images [56], from the model. Various models including diffusion [56, 57], transformers [44, 55, 58], and GANs [59–61] are vulnerable to this attack. For example, since memorized samples tend to appear with disproportionately high frequency and confidence scores, attackers can prompt the model to generate a large number of outputs, rank them based on these metrics, and then examine them for memorization. Extracted text sequences can contain sensitive information such as names, phone numbers, and email addresses [44, 55], while extractable images contain identifiable faces and company logos [56].

Other privacy attacks: Generative models are also vulnerable to additional privacy risks, including attribute inference and linkability attacks. Attribute inference attacks [10, 20, 62–64] aim to uncover a specific sensitive attribute of the model's training data such as its composition. In contrast, a linkability attack [65, 66] concerns with whether an adversary can determine if two records originate from the same underlying entity.

2.3 Privacy Risks of Network Traffic

Fingerprinting attacks are among the most widely studied and prevalent for network traffic. Leveraging packet sizes, timing patterns, TTL values, and window sizes, adversaries can infer device types [15, 16], operating systems [21, 22], and services being used [67, 68]. Under attribute inference attacks [17, 23], attackers can infer sensitive attributes about training data such as demographic of its users, their age group and career stage. Additionally, some attacks [1, 24, 25]

attempt to deanonymize networks and reconstruct network topologies from anonymized traffic data.

Despite various privacy attacks on network traffic, privacy risks in synthetic network traffic remain largely unexplored. Jin et al. [14] provide early evidence that MIA at the traffic-source level is feasible. However, their work focuses solely on MIA, leaving other threats, such as fingerprinting and attribute inference, unexplored. This leaves a critical gap in understanding the privacy risks of synthetic network traffic and how to mitigate them.

2.4 Privacy Risks Mitigation

Various privacy risks mitigation techniques have been developed to protect network traffic privacy. Anonymization methods [69–71], such as prefix-preserving and reversible pseudonymization, mask identifiers like IP and MAC addresses to prevent recovery of original values. Other approaches protect sensitive fields, including port numbers, flow sizes, and packet inter-arrival times, by reducing metadata granularity [72], adding noise [73, 74], or modifying packet traces through traffic shaping [75–77].

Despite extensive work on privacy protection for real network traffic, synthetic network traffic remains largely unprotected. Existing generators either lack privacy safeguards [11, 12] or apply limited protections, such as removing IP addresses [4] or adding differential privacy noise to protect against MIA [3, 13]. Moreover, none evaluate or mitigate network traffic–specific privacy attacks. This lack of comprehensive privacy evaluation and mitigation poses significant security challenges for synthetic network traffic generation.

3 Methods to Evaluate Privacy & Utilities

Section 3.1 describes our methods for evaluating privacy risks through MIA, data extraction, and network-specific privacy attacks, along with the rationale for these attacks. Section 3.2 presents the mitigation techniques we evaluate and the downstream utility assessments used to measure their privacy–utility tradeoffs.

Not all privacy attacks apply to every generative model due to differences in model architectures and outputs. For example, NetDiffusion removes all network identifiers from the training data, preventing identifier-based attacks. Similarly, TrafficLLM generates only the first packet of each flow rather than complete flows, making attribute-based network-specific attacks infeasible. Accordingly, we evaluate only the privacy attacks applicable to each model. Table 2 summarizes the attacks conducted for each model.

3.1 Privacy Attack Methods

*3.1.1 Membership Inference Attack (MIA)*We conduct MIA at the PCAP level, the common generation unit for network traffic models. To launch MIA, we curate signals (features) for training binary classifiers that predict whether a sample belongs to the training set. Under black-box MIA, we assume the adversary has no access to model internals, such as parameters or gradients, and does not train shadow models. However, the adversary can observe model outputs and token log probabilities [44]. For NetDiffusion, we use the pixel-wise L_2 reconstruction error between the input and generated images. For NetSSM and TrafficLLM, we compute perplexity using token log probabilities from the first packet of the input PCAP. For NetShare, which generates tabular flow records, we extract features from generated outputs, including the total number of packets, counts of unique source and destination IPs and ports, and the mean and standard deviation of continuous fields such as packet length and TTL.

	MIA	Data extraction	Network-specific Identifiers	Attributes	Topology
NetShare	✓	✗	✓	✓	✓
NetDiffusion	✓	✓	✗	✓	✗
TrafficLLM	✓	✓	✓	✗	✗
NetSSM	✓	✓	✓	✓	✓

Table 2: Privacy attacks conducted for each generative model. ✓ indicates an attack was conducted; ✗ indicates "not conducted" because of different data encodings and model design choices.

Under white-box MIA, where the attacker can access model parameters and gradients, we use the average gradient norm of model parameters on the input image for NetDiffusion. For TrafficLLM and NetSSM, we use the average gradient norm computed on tokens generated from the first packet of the input PCAP. We do not conduct white-box MIA on NetShare because its generator and discriminator operate on preprocessed encoded representations rather than raw traffic data. Since the encoded input dimensionality depends on the dataset-specific encoding process, models trained on one PCAP trace are incompatible with others, preventing direct reuse of model internals for membership inference.

After extracting signals from the target holdout set (non-members) and shadow training set (members), we train classifiers to predict membership status. We evaluate several classifiers, including neural networks, support vector machines (SVMs), and XGBoost, and report results from the best-performing model. Following standard MIA evaluation practice [78–80], we report the true positive rate at a false positive rate of 0.01 (TPR at FPR$\leq$0.01). Figure 1 illustrates the construction of the train, shadow train, and target holdout sets, while Figure 2 shows the overall MIA pipeline.

*3.1.2 Data Extraction Attack*Inspired by Nasr et al. [55], who define a successful extraction as one where the model outputs text containing a substring of at least 50 tokens that appears verbatim in the training set, we adapt this approach to the networking context as follows. First, we define the "generation prompt" as the input required by each model to produce

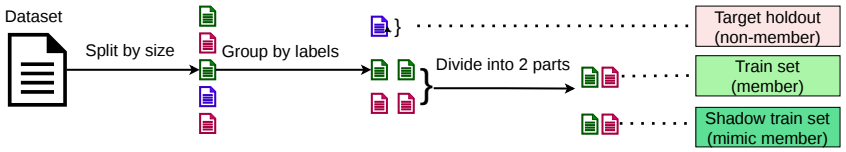


Figure 1: Pipeline of splitting each dataset into train set, shadow train set and target holdout.

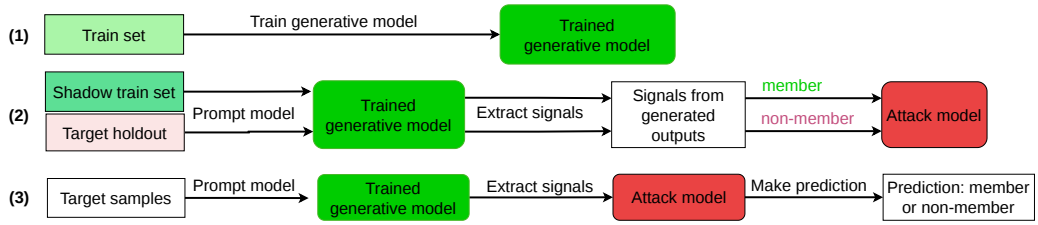


Figure 2: Pipeline of Membership Inference Attacks (MIA). (1) The data owner trains the generative model on the train set and makes it available as a service to everyone. (2) An attacker queries the generative model to extract signals and trains a binary attacker model to predict membership. (3) To predict membership of a target sample, an attacker first queries the generative model to extract signals, then predicts membership using the trained attacker model.

synthetic traffic. For NetDiffusion, this is a combination of a training image and a descriptive text prompt specifying the traffic type. For NetSSM and TrafficLLM, it is the traffic type label present in the training data. For NetShare, it is the raw network traffic PCAPs.

Second, we compare each generated sequence against the training data to identify verbatim matches. We define a generated sequence as *extractable* if it contains a *contiguous sequence of at least 10 tokens* that appears verbatim in the training data (we identify contiguous matches at the token level, rather than in the parsed header fields). Because different generative models operate over different representation units, we define a "token" as the native unit used by each model to learn and generate synthetic traffic. For example, in NetDiffusion, a token corresponds to a bit, so extractability requires memorization of a contiguous sequence of 10 bits (1.25 bytes). In NetSSM, a token corresponds to a byte, yielding a threshold of 10 bytes, while in TrafficLLM, a token corresponds to a hexadecimal digit, corresponding to 10 hex digits (5 bytes).

To measure the success of a data extraction attack on a model, we introduce the metric *extractable rate*, defined as *the percentage of generated outputs that constitute a successful extraction.*

*3.1.3 Network-Specific Attack*Network traffic contains diverse information that varies in sensitivity. Network identifiers such as MAC addresses and IP addresses are highly sensitive, as they can reveal the location, device type, and even the identity of users. Other features, such as Time To Live (TTL) and Type of Service (ToS), while less sensitive than network identifiers, can be used to generate fingerprints that infer the operating system or application type. Furthermore, overall traffic patterns can be collectively analyzed to reconstruct network topology, exposing critical network infrastructure. To address these diverse privacy concerns, we categorize this sensitive information into three types: network identifiers, network properties, and network topology, and we evaluate each separately.

Network Identifiers: While network identifiers are considered "personal information" under major privacy frameworks, including GDPR and CCPA, and may be subject to anonymization requirements, certain entities such as network operators, ISPs, and organizations exempt under specific jurisdictions, may lawfully retain this information for legitimate operational purposes, including security monitoring, network management, and attack detection. Therefore, identifier leakage in synthetic traffic therefore remains a significant practical concern. Inspired by the privacy metrics of Lukas et al. [81] for measuring PII leakage, we adopt two complementary metrics, Precision and Recall, to quantify identifier leakage. *Recall* measures how much of the training dataset's network identifiers are at risk of extraction. *Precision* measures an attacker's confidence that a generated network identifier appears in the training dataset.

$$Recall = \frac{|\{\, i \mid i \in \mathcal{G} \cap \mathcal{T} \,\}|}{|\{\, i \mid i \in \mathcal{T} \,\}|}$$

$$Precision = P(i \in \mathcal{T} \mid i \in \mathcal{G})$$

where $\mathcal{G}$ denotes the set of identifiers generated, and $\mathcal{T}$ denotes the set of identifiers present in the training data.

Network properties: Based on features used in fingerprinting and attribute inference attacks [17, 21–23], we consider the following non-identifiers as sensitive network properties: Time To Live (TTL), IP Identification (IP ID), Type of Service (ToS), Window Size, Control Flags, TCP Data Offset, flow size, and packet size. Since ground truth for fingerprinting attacks is unavailable, we evaluate leakage by comparing

the distributions of these features between training and generated data using Earth Mover's Distance (EMD) [82] [1]. A smaller distance indicates higher leakage risk.

Network topology: Network topology refers to the structure of how devices are connected and communicate in a network, including link-level connections (using MAC addresses) and network-level routing (using IP addresses). This information can expose relationships among devices and network segments [83], allowing attackers to identify critical nodes and potential attack paths [84, 85]. To measure topology leakage, we employ a graph-based comparison between training and generated data. We define a communication flow as an exchange of network traffic (unidirectional or bidirectional) between a pair of addresses (IP or MAC).

To evaluate topology leakage, we select the top 100 communication flows ranked by total bytes transferred and constructs non-directed graphs where nodes represent IP or MAC addresses and edges encode observed communication relationships. Since the flows in this setting are large, persistent transfers that dominate network by volume despite constituting a negligible fraction of total flow count, they likely encode inter-node dependency relationships, bottleneck locations, and routing hierarchies, making their leakage a disclosure of the load-bearing skeleton of the underlying network topology [86]. Additionally, we also evaluate under a setting where the topology comprises of both large and small flows. Refer to Appendix C for the setup and results.

Topology similarity is evaluated using three metrics: (1) node overlap—the proportion of addresses common to both graphs, (2) edge overlap—the proportion of shared communications, and (3) node connectivity distribution—the similarity in the number of connections per node across both topologies (measured using EMD). Note that cross-flow correlation attacks, while outside our current threat model, could amplify topology leakage by exploiting relationships across flows and are an important direction for future work.

3.1.4 Why do These Attacks Matter? We focus on membership inference, data extraction, and network-specific privacy attacks because they correspond to the main questions that network operators and data owners are concerned about when releasing synthetic traffic. Industry frameworks such as the OWASP ML Security Top 10 [87] and NIST AI guidance [88] recognise MIA and data extraction as AI privacy risks that can lead to confidentiality breaches, regulatory exposure, and reputational harms.

First, MIA captures *participation privacy* and is widely adopted as a standard way to quantify privacy risk in machine learning [46]. In the networking context, MIA reveals whether traffic from a specific user, device, service, or organization was used for training, allowing attackers to infer sensitive network activity or the presence of specific services or devices.

Second, data extraction attacks measure *content memorization*. Prior work has shown that generative models can regurgitate individual training records [44, 55]. In network traffic, these attacks test whether a model reproduces training packet sequences when queried. For example, extracting packets from an internal service could reveal its port, protocol, and communication pattern, enabling attackers to identify and target hidden services.

Third, our network-specific attacks target *semantic leakage* unique to network traffic. By separately measuring leakage of network identifiers (such as IP and MAC addresses), fingerprinting-relevant fields (such as TTL, window size, or flags), and communication topology, these attacks capture concrete harms such as re-identifying a host from its IP address, inferring its OS or software stack from fingerprinting fields, or identifying otherwise hidden internal services from communication patterns.

Together, these three attack families capture complementary privacy risks in synthetic network traffic: training-set participation (MIA), trace memorization (data extraction), and network-semantic leakage (network-specific attacks). We therefore argue that they provide a meaningful core for evaluating privacy risks in traffic generative models.

In this study, we focus on privacy risks in *multi-flow* network traffic, where packet captures (PCAPs) contain multiple concurrent flows, as this better reflects real-world network conditions and often introduces additional privacy risks compared to single-flow traffic. Further motivation for this design choice is provided in Section 4.1.

3.2 Privacy Risks Mitigation Methods

Inspired by privacy risks mitigation techniques for real network traffic [69–71, 73, 74], we explore several mitigation strategies and evaluate their privacy-utility tradeoffs on synthetic traffic. For each strategy, we apply mitigation on the traffic generators' training data.

3.2.1 Anonymization-Based Strategies To protect network identifiers and network topology, we investigate three anonymization strategies with varying levels of restrictiveness.

- **Complete Anonymization (CA):** In this most restrictive anonymization approach, each network identifier (IP addresses and MAC addresses) is replaced with a single constant value. This approach effectively eliminates all identifier-based privacy attacks, including network topology reconstruction.
- **Pseudonymization (PS):** A less restrictive alternative is pseudonymization where each unique network identifier in the training data is consistently replaced with a distinct

[1] EMD sums the absolute differences between the cumulative distributions across bins and normalizes the result by the maximum possible EMD ($n-1$), yielding a value in $[0, 1]$.

randomly generated synthetic identifier. This one-to-one mapping preserves network structure and communication patterns while obscuring the original identifiers.

- **Prefix-preserving (PP):** For this approach, the last k bits of each network identifier are set to zero (k=8 bits for IP addresses, k=12 bits for MAC addresses), while the rest remain intact, allowing the hierarchical prefix information to be preserved. For IP addresses, subnet structure is retained while host identities is obscured. For MAC addresses, the organization unique identifier that identifies the manufacturer is preserved while device-specific bits are removed.

*3.2.2 Noise-Based Strategies*Differential Privacy (DP) noise is a privacy risk mitigation technique where calibrated random noise is deliberately injected into data, queries, or model updates, ensuring that it is statistically impossible for an adversary to detect whether any single data point was included or excluded in the dataset. Since most network traffic generative models do not currently support adding DP noise directly to the model (e.g., via DP-SGD), we apply DP-noise to the training data, as in previous works [13]. Rather than perturbing every header field, we specifically target the sensitive network properties including TTL, IP ID, ToS, Window Size, IP packet length. We do not add noise to fields such as IP addresses or MAC addresses. We allocate an equal privacy budget across all targeted properties and add noise simultaneously for all selected properties. We then apply clipping and rounding operations to ensure the perturbed values remain within valid ranges and conform to the appropriate data types.

*3.2.3 Privacy-Utility Tradeoffs Evaluation*We measure utility using two metrics: fidelity and downstream task performance. To assess privacy implications, we select key metrics from Section 3.1 covering all three attack types.

Fidelity metrics. Fidelity measures how well synthetic traffic resembles real-world traffic. Following the fidelity metrics used in NetShare [3], we measure distributional distance between training and generated datasets using Earth Mover's Distance (EMD) across five key distributions. *An increase in EMD indicates a decrease in fidelity* since the generated data diverges further from the training distribution, while *a decrease in EMD indicates an increase in fidelity*. The five evaluated distributions are:

- **SA/DA:** Relative frequency ranking of source and destination IP addresses (from most to least frequent)
- **SP/DP:** Distribution of source and destination port numbers
- **PR:** Relative frequency of IP protocols (e.g., TCP, UDP, ICMP)

Traffic Dataset	#Pcaps	#Flows	#Packets	#IPs	#MACs	#Labels
Service recognition (SR)	660	2,363	1,313,372	207	21	11
IoT (IoT)	521	9,023	1,007,501	1324	48	9
USTC TFC 2016 (USTC)	501	183,570	994,174	116620	15	13
CIC DoHBrw 2020 (CIC)	480	10,758	838,140	2330	7582	8
VNAT (VNAT)	1103	14,646	2,118,603	52	51362	19

Table 3: Overview of five network traffic datasets.

Downstream task performance. Downstream task utility of synthetic traffic is measured using downstream classification accuracy. In the multi-flow setting, we classify PCAPs using PCAP-level labels. In the single-flow setting, we split each PCAP into flows and classify individual flows using flow-level labels. Since flow-level labels are available only for the IoT and SR datasets, to ensure fair comparison between multi-flow and single-flow settings, we evaluate downstream performance only on these two datasets.

For the IoT dataset, the multi-flow task is device activity classification (e.g., Alexa playing music), while the single-flow task is device-type classification, which is important for anomaly detection and security monitoring [89, 90]. For the SR dataset, both multi-flow and single-flow tasks classify service types (e.g., video conferencing or streaming), which are important for QoS provisioning and network resource allocation [91].

Similar to Jiang et al. [4] and Chu et al. [11], we also use packet header-level features (excluding identifier-based features such as IP addresses) determined via nPrintML [92], as features to classify each PCAP. [2] We train two XGBoost classifiers: one on synthetic traffic generated when the generator's training data is unmitigated and one on synthetic traffic generated when training data is mitigated. We evaluate these two classifiers on the same real-world held-out test set (shadow train set). The classification accuracy difference of these two models quantifies the utility cost of applying mitigation techniques.

Privacy metrics. Privacy impact of mitigation techniques is measured using the follow metrics. To measure changes in MIA success, we measure changes in TPR at FPR ≤ 0.01 (Δ TPR at FPR ≤ 0.01). For data extraction attack, we use changes in extractable rate (Δ extractable rate). Lastly, for network-specific attacks, we use (1) changes in precision score and (2) changes in recall score for source IP and (3) changes in Earth Mover's Distance (Δ EMD) between training and generated data averaged across all sensitive network properties (refer to Section 3.1.3). Since lower EMD indicates more successful attacks (closer distributional match enables easier property inference), while higher EMD indicates improved privacy protection, to maintain consistency with other metrics where positive changes indicate improved privacy, we

[2]We use packet header–level features instead of flow-level features because anonymization techniques such as CA can remove the information needed to identify flows.

report the change in network-specific attack effectiveness as the negative of the change in EMD (i.e., -ΔEMD).

We investigate privacy–utility tradeoffs for both multi-flow and single-flow generation to understand how mitigation techniques affect different traffic generation tasks. Privacy is evaluated across all five datasets, while utility, including both fidelity and downstream task performance, is assessed only on the IoT and SR datasets.

4 Experimental Setup

This section describes the target to be generated in our experiments, the dataset preparation process, and the details of synthetic traffic generation.

4.1 Generation Target

In this work, our main target for generation for privacy evaluation is *multi-flow PCAPs*. Unlike single-flow PCAPs that capture communication between two entities, multi-flow PCAPs contain multiple concurrent flows, each defined by a 5-tuple (source IP, destination IP, source port, destination port, protocol). Multi-flow traffic generation is valuable for practical applications including network traffic analytics, monitoring, and simulating realistic environments to test network algorithms and protocols. However, it presents unique technical and privacy challenges that make it particularly important to study.

From a technical perspective, multi-flow generation is more complex than single-flow generation. While single-flow generation only needs to model the temporal dynamics within one communication session (intra-flow patterns), multi-flow generation must additionally capture the relationships and dependencies between different flows (inter-flow patterns), such as timing correlations and bandwidth sharing across concurrent connections.

From a privacy perspective, multi-flow traffic exposes more sensitive information than single-flow traffic. Beyond individual communication details, multi-flow data reveals network-wide communication patterns, entity relationships, and network topology—information that is highly sensitive and valuable to adversaries. For example, flow co-occurrence patterns can serve as effective signals for membership inference and topology reconstruction attacks, and can further enable the construction of user, service, or device traffic fingerprints. These elevated privacy risks make multi-flow traffic generation a critical area for privacy evaluation.

Given both the practical importance and heightened privacy concerns of multi-flow traffic generation, we focus primarily on this scenario throughout our evaluation.

4.2 Dataset Preparation

We use five datasets collected from different networks and application domains. The IoT dataset [93] contains traffic from various IoT device activities. The Service Recognition (SR) [4] and VNAT [94] datasets capture traffic from different service types, including streaming, conferencing, and social media. The USTC TFC 2016 (USTC) [95] and CIC DoHBrw 2020 (CIC) [96] datasets provide network traffic from various attack scenarios. [3] Refer to Table 3 for the final dataset size.

Furthermore, since packet captures (PCAPs) vary widely in length — with some exceeding 100,000 packets — we split each PCAP into smaller segments of up to 2,000 packets and randomly select at most 20 segments from each original PCAP. Additionally, since some labels contain significantly more PCAPs than others, we limit each label to a maximum of 60 PCAPs to ensure balanced representation across labels. Table 3 provides details about each dataset after this curation process.

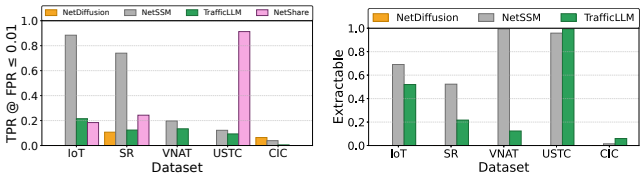
For each dataset, we first group PCAPs by label. We then randomly select two labels and set aside all PCAPs from those labels as target holdout (non-member). The PCAPs associated with the remaining labels are split evenly (50/50) into two parts: one part is used to train the generative models (train set), and the other serves as the shadow train set (mimic members). Refer to Figure 1 for how each dataset is split.

4.3 Synthetic Traffic Generation

We train NetDiffusion by fine-tuning Stable Diffusion v1.5 using LoRA (Low-Rank Adaptation) with a batch size of 4. We train until the average loss over the most recent 100 epochs falls below 0.02 (an epoch is one complete pass over the training set). This empirically chosen threshold indicates a clear loss plateau across most datasets. For NetSSM, we use a batch size of 1 and train until the loss stabilizes over the most recent 4 epochs. Similarly, TrafficLLM is fine-tuned from ChatGLM2-6B with a batch size of 1 and trained until the loss plateaus over the most recent 4 epochs. For NetShare, we adopt the default training configuration without Differential Privacy (DP). Since GAN training involves adversarial optimization between a generator and a discriminator, both losses tend to oscillate rather than converge monotonically; therefore, we train until the losses stabilize and fluctuate around a consistent level.

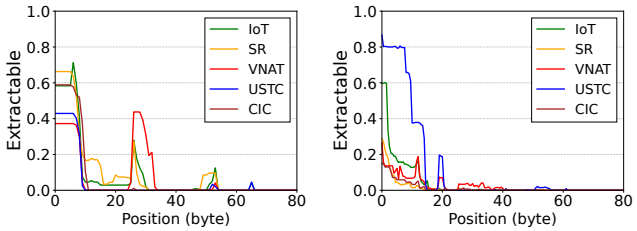
After training traffic generative models, we generate synthetic traffic as follows. For NetDiffusion, we generate 20 output images for each input image (where each image represents a multi-flow PCAP file). We select the output closest to the input based on pixel-wise L2 distance and apply the post-generation correction process to the selected image. For NetSSM and TrafficLLM, we generate 30 output sequences per traffic label. This matches the approximate number of PCAPs per label in the training data, ensuring balanced sample representation across labels. For NetShare, we use the

[3] Note that we use only a subset of labels from each dataset rather than the full dataset, as some generative models are computationally expensive to train. This also ensures comparable dataset sizes across experiments.



(a) MIA success (b) Data extraction success

Figure 3: MIA and data extraction success across models and datasets.



(a) NetSSM (b) TrafficLLM

Figure 4: Extractable rate across different byte positions for NetSSM & TrafficLLM. Higher extractable rate is observed at byte positions 1-9 and 26-33 for NetSSM.

default setting where the number of flows generated is approximately equal to the number of flows in the input PCAP.

5 How successful can each privacy attack be under the most challenging conditions for an attacker?

To address RQ1, we begin with a conservative setting in which the attacker is restricted to black-box access, and all generative models are trained for a single epoch. Under this setting, we observe that all models are under-trained. Refer to Section 6 for results when models are trained for extended epochs.

5.1 MIA Success Rate

Finding 1: MIA can be highly successful against NetSSM and NetShare on certain datasets.

Even under the setting where attacker only has black-box access and the model is under-trained, MIA is highly effective against NetSSM and NetShare on certain datasets. For NetSSM, TPR at FPR ≤ 0.01 can be as high as 0.88 on IoT dataset and for NetShare, TPR at FPR ≤ 0.01 reaches over 0.9 on USTC dataset. MIA is moderately effective against TrafficLLM, but considerably less effective against NetDiffusion (Figure 3a). We hypothesize that these differences arise from differences in data encoding and model architecture. Compared to bit-level encoding in NetDiffusion, byte-level or hexadecimal representations in NetSSM and TrafficLLM may provide more informative representations of network traffic. Their language-based architectures may also better capture the sequential structure of network traffic than NetDiffusion. As such, MIA attacks tend to be more successful against NetSSM and TrafficLLM than NetDiffusion. However, the signals available to the MIA attacker also differ across models due to their different input/output representations. Thus, cross-model differences in MIA success should be interpreted cautiously, as they do not isolate the effect of architecture alone.

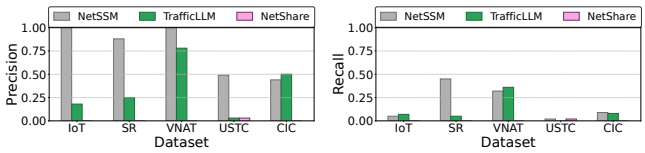
Furthermore, the effectiveness of MIA varies substantially across datasets. For example, MIA achieves high success rates against NetShare on the USTC dataset, but performs poorly on other datasets. Similarly, MIA is highly successful against NetSSM on the IoT and SR datasets, while showing limited effectiveness on the remaining datasets. One possible explanation is the distribution difference between the train and holdout set for the features used by MIA attack. For instance, if the first packets of PCAPs from the train set closely resemble those from the holdout set, TrafficLLM and NetSSM may struggle to distinguish between the train and holdout samples. Likewise, if flow-level features, such as the number of source IPs or packet counts, is highly similar, NetShare may fail to effectively differentiate between them. Additional evidence supporting this explanation is provided in the Appendix Figure 12 and 13.

5.2 Data Extraction Attack Success Rate

Finding 2: Data extraction attack is more successful against autoregressive models than diffusion models. Categorical/identifier data are more likely to be memorized than continuous data.

Among the three packet-level generative models, NetDiffusion is not vulnerable to data extraction attacks. This aligns with the design choice in NetDiffusion: representing traffic as images at bit granularity removes explicit sequence structure, which likely makes verbatim packet-level memorization harder to sustain. In contrast, the extractable rate is notably high (≥ 0.4) for both NetSSM and TrafficLLM which are autoregressive models. Refer to Figure 3b for more details about data extraction attack success. It is important to note that our extraction analysis uses each generator's native output representation and token granularity, rather than a common unit. Thus, extraction rates are not directly comparable across generators.

We next analyze which byte positions in packet headers are most vulnerable to extraction. A position is defined as *extractable* if the next 10 contiguous tokens starting at that location appear identically in the training data at the same offset (Figure 4). For NetSSM, extractable rates are consistently high at byte positions 1–9 and 26–33 across datasets, indicating these regions are more predictable and susceptible to memorization. These positions correspond primarily to identifier fields: bytes 1–9 map to Ethernet headers



(a) Precision (b) Recall

Figure 5: Precision and recall scores for source IP addresses across different models and datasets.

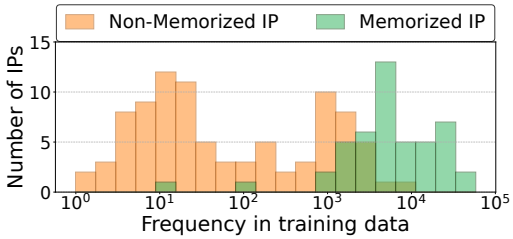


Figure 6: Distribution of memorized and non-memorized source IPs as a function of their frequency in the training data for dataset SR (Total number of IPs = 227). Source IPs that appear more frequently in the training set are more likely to be memorized and reproduced in the generated output.

(source/destination MAC addresses and EtherType), while 26–33 map to network-layer fields such as source/destination IP addresses and related padding. In contrast, mid-range positions (e.g., 14–25), which include more continuous-valued fields such as packet size and TTL, show lower extractability due to higher variability and lower repetition. Overall, extraction success is concentrated in categorical header fields, while continuous fields are substantially less vulnerable.

We also observe that TrafficLLM exhibits a similar extractability pattern to NetSSM, with consistently high extractable rates at byte positions 1–15 (corresponding to bytes 1–20 in TrafficLLM, since each token represents a hexadecimal digit). However, unlike NetSSM, it does not show similarly high extractability at positions 26–33. This difference could be due to tokenization granularity: NetSSM operates at the byte level, whereas TrafficLLM uses hexadecimal digit tokens, resulting in longer sequences for the same information and making exact memorization more difficult.

*5.2.1 Network identifiers leakage*Source and destination IP addresses show substantial leakage in both precision and recall metrics (Figure 5 for source IPs; Figure 14 in the Appendix for destination IPs). This leakage is most severe for NetSSM, followed by TrafficLLM while not significant for NetShare. We hypothesize that precision and recall scores is low for Netshare because identifiers like source IP are encoded into vector embedding of lower precision, which reduce the risk of exact memorization of such fields. For NetSSM, precision scores on source IP exceed 0.8 on three of five datasets, indicating a high likelihood that any generated identifier appears in the training set. We also observe that recall scores is particular high for the SR and VNAT datasets

Feature	NetDiffusion	NetSSM	TrafficLLM	NetShare
Time to Live (TTL)	0.36	**0.10**	0.22	0.32
Identification (ID)	0.26	**0.08**	0.18	0.27
Type of Service (ToS)	0.17	**0.06**	0.11	0.27
TCP Window Size	0.14	**0.12**	0.25	-
TCP Control Flags	**0.15**	**0.15**	0.16	0.30
TCP Data Offset	0.16	0.28	**0.14**	0.50
Flow Size	-	**0.05**	-	0.13
Packet Size	0.14	**0.12**	-	0.28

Table 4: Normalized EMD between the original and generated distributions for each sensitive feature. Bolded values indicate the lowest distance (more leakage) for each feature.

Traffic Dataset	Degree distribution distance	
	MAC-based	IP-based
IoT	**0.04**	**0.01**
SR	**0.05**	**0.04**
VNAT	0.17	**0.04**
USTC	0.19	**0.01**
CIC	0.40	**0.00**

Table 5: Normalized EMD of node degree distributions across training and generated topologies.

(>0.3). We hypothesize that, since these datasets contain relatively fewer IP addresses than the other datasets, achieving high recall on these datasets is more tractable compared to datasets like USTC which contain thousands of unique IPs. Refer to Table 3 for number of identifiers in each dataset.

Furthermore, we observe that identifiers appearing with higher frequency in training data are more likely to be memorized (refer to Figure 6 for an example). For example, source IPs appearing more than 1,000 times in the training data show substantially higher memorization rates compared to those with lower frequencies. This pattern reveals a critical privacy vulnerability: frequently occurring network identifiers, which often correspond to important servers, major clients, or central network infrastructure, face disproportionately higher exposure risks.

*5.2.2 Network properties leakage*The EMD between training and generated data can be as low as 0.10 or less for four key properties (TTL, ID, Flow Size, and ToS), indicating that these features can remain highly useful for fingerprinting attacks. Note that the EMD for Flow Size is unavailable for NetDiffusion, which removes IP addresses used for flow identification, and for TrafficLLM, which generates only the first packet. Across all four generative models, NetSSM exhibits the greatest privacy leakage (lowest EMD). For the other three models, leakage varies by feature: some features show substantially higher distances while others remain closely matched to the training distribution. NetShare, a flow-based generative method, exhibits the largest distribution distance overall, suggesting relatively better privacy preservation for sensitive properties compared to packet-based generative models. Refer to Table 4 for details.

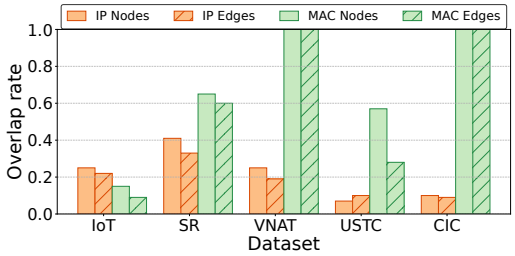


Figure 7: Topology overlap rate (nodes and edges) for IP-based and MAC-based communications across different datasets for NetSSM.

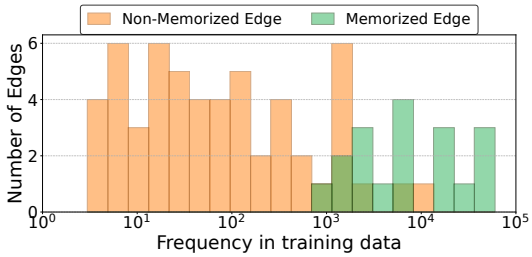
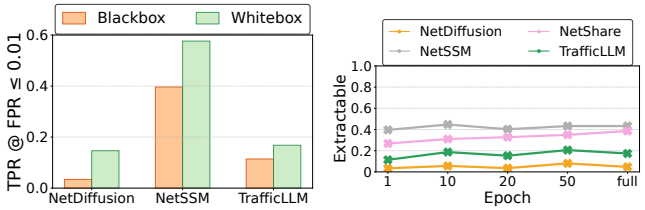


Figure 8: Distribution of memorized and non- memorized edge as a function of their frequency in the training data for dataset SR. Edges that appear more frequently in the training set (belong to larger flows) are more likely to be memorized and reproduced in the generated output.

Finding 3: High traffic volume and highly connected nodes are likely to be exposed in generated topologies.

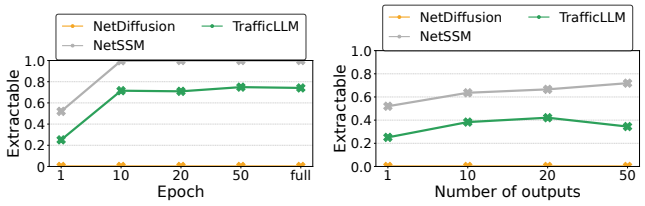
*5.2.3 Network topology leakage*NetShare demonstrates minimal network topology leakage with very low node and edge overlap rates, which is non-surprising as it does not leak network identifiers (refer to Figure 5). In contrast, traffic generated by NetSSM exhibits substantial topology leakage (see Figure 7 for details). For IP-based topologies, NetSSM achieves node overlap rates up to 0.4 and edge overlap rates up to 0.35. For MAC-based topologies, overlap rates reach 1.0 for both nodes and edges, indicating complete topology reconstruction. The low degree-distribution EMD further suggests that the generated topologies closely preserve the training graph structure.

These high topology overlap rates demonstrate that NetSSM memorizes and reproduces significant portions of the network topology, including key entities and their communication patterns. Manual inspection shows that overlapped IP nodes are high-volume endpoints, such as external CDN, cloud infrastructure servers and a substantial number of private IP addresses linked to high-volume flows. For example, in the SR dataset, we identify multiple private IPs associated with personal devices generating large volume of traffic for applications such as Zoom and Google Meet.



(a) White-box vs black-box (b) Number of training epochs

Figure 9: Impact of different experimental conditions on MIA effectiveness.



(a) Number of training epochs (b) Number of outputs

Figure 10: Impact of different experimental conditions on data extraction success.

6 Which factors determine the success of privacy attacks?

6.1 Factors Affecting MIA

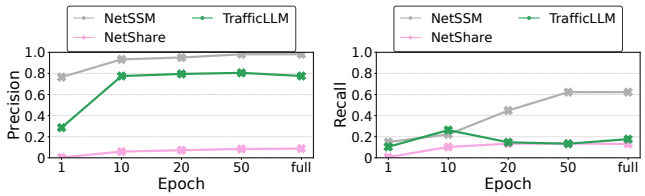
Finding 4: White-box access can substantially increase MIA effectiveness against NetSSM and NetDiffusion.

White-box access increases TPR at FPR ≤ 0.01 by an average of 0.15 for NetDiffusion and NetSSM, indicating that attacker access level is a significant factor influencing MIA success for these models. In contrast, MIA success for TrafficLLM remains largely unchanged between black-box and white-box settings, suggesting that access level has little influence on attack effectiveness for this model. See Figure 9a for details.

Although training generative models for more epochs can slightly increase MIA success for some models, the overall effect is modest. For NetSSM, TPR at FPR ≤ 0.01 is already about 0.40 after one epoch and increases only marginally to around 0.45 at full convergence. For NetShare, TPR rises from 0.25 at epoch 1 to approximately 0.40 as training completes. TrafficLLM shows relatively stable performance, with TPR fluctuating between 0.10 and 0.20 across epochs, while NetDiffusion remains resistant to MIA, maintaining TPR below 0.07 throughout training. Overall, although extended training can increase MIA success, the gains are limited, suggesting that models already capture membership signals early in training (Figure 9b).

6.2 Factors Affecting Data Extraction

Training for more epochs substantially increases data extraction vulnerability in NetSSM and TrafficLLM. From 1 to 10



(a) Precision **(b) Recall**

Figure 11: Impact of the number of training epochs on precision and recall scores for source IP address.

epochs, extractable rates rise from 0.65 to 1.0 for NetSSM and from 0.27 to 0.73 for TrafficLLM, then largely saturate. In contrast, NetDiffusion remains near zero across all stages, indicating strong resistance to extraction. Overall, longer training increases extraction risk, particularly for under-trained NetSSM and TrafficLLM models (Figure 10a).

Increasing number of generated outputs can also increase data extractability, although to limited extent. For example, for NetSSM, data extractability can increase from 0.52 to 0.73 when number of generated outputs increase from 1 to 50. TrafficLLM increases more moderately, from approximately 0.27 to a peak of around 0.42 at 20 outputs, before slightly declining to approximately 0.35 at 50 outputs. In contrast, NetDiffusion remains near 0 regardless of the number of outputs, demonstrating consistent resistance to data extraction. Refer to Figure 10b for details.

6.3 Factors Affecting Network-specific

NetSSM exhibits high vulnerability to source IP extraction. Starting at epoch 1, its precision score already reaches 0.78 and steadily increases to near-perfect levels by epoch 20. Its recall score also increases substantially, from 0.13 at epoch 1 to around 0.63 at epoch 50. This suggests that additional training epochs not only improve memorization accuracy but also increase the number of identifiers memorized by the model. The precision score for TrafficLLM also increases with additional training epochs, rising from 0.27 at epoch 1 to around 0.80 by epoch 10 before stabilizing. In contrast, although both the precision and recall scores for NetShare increase with additional training epochs, they remain low overall, reaching only around 0.10 for precision and 0.18 for recall, indicating strong resistance to source IP extraction. Overall, these results suggest that the degree of model fitting affects not only precision but also recall across all three models. Refer to Figure 11 for more details.

We also found that the number of generated output does not significantly impact network identifier meorization. Refer to Appendix Figure 15 for more details.

7 What is the privacy-utility tradeoff when applying privacy mitigation techniques?

To address RQ3, we evaluate how privacy mitigation strategies affect both privacy risks and downstream utility. We focus on NetSSM, as it exhibits the strongest leakage across the studied attacks, and evaluate it under both multi-flow and single-flow generation settings. To ensure a fair comparison between these settings, we evaluate mitigation strategies only on the SR and IoT datasets, where labels are available for the single-flow setting.

As described in Section 3.2, we apply three anonymization techniques: Complete Anonymization (CA), Pseudonymization (PS), and Prefix-preserving (PP). For DP-noise mitigation, we consider two privacy budgets, $\epsilon = 0.1$ (stronger privacy) and $\epsilon = 1$ (weaker privacy). We evaluate the impact of these mitigation methods on privacy across five datasets, while the impact on utility is assessed only on the IoT and SR datasets.

7.1 Impact of Anonymization Techniques

Finding 5: Anonymization techniques significantly reduce MIA and data extraction success.

Across both multi-flow and single-flow settings, all three anonymization strategies eliminate the risks of data extraction and network identifier attacks. The extractability dropping to 0 as network identifiers are removed indicate that header-level memorization is strongly tied to raw identifiers. Interestingly, anonymization also substantially reduces MIA success in both the single-flow (by 0.24–0.37) and multi-flow (0.23–0.44) settings. This likely occurs because the model is trained on anonymized tokens and therefore learns representations aligned with the anonymized distribution rather than the original data. As a result, it produces weaker membership signals with respect to the original data, making training samples harder to distinguish from non-members. Anonymization techniques, however, do not protect sensitive network properties, as reflected by near-zero changes in EMD across all methods (CA, PP, and PS).

With respect to data utility, anonymization techniques reduce both fidelity and downstream accuracy in the multi-flow and single-flow settings. Compared to DP-noise mitigation techniques, anonymization tends to cause a larger drop in fidelity but a smaller reduction in downstream accuracy. One possible explanation is that anonymization mainly alters network identifier distributions which is used to calculate fidelity, while preserving non-identifier fields that are used for downstream tasks.

7.2 Impact of Noise-addition Techniques

Finding 6: DP noise protects against network properties inference attack but not other privacy attacks.

In both the single-flow and multi-flow settings, adding DP noise has negligible impact on data extraction success as well as on the precision and recall of source IP address

Generation task	Mitigation strategy	Detail	Privacy attack (↓ more private) MIA	Extraction	Network EMD	Network Precision	Network Recall	Utility (↑ more utility) Fidelity	Utility Accuracy
Multi-flow	No-mitigation		0.81	0.66	0.10	0.94	0.18	0.12	0.64
	Anonymization	Complete anonymization (CA)	↓ -0.23	↓ **-0.66**	↓-0.01	↓**-0.94**	↓**-0.18**	-0.05	-0.03
		Pseudonymization (PS)	↓**-0.44**	↓ **-0.66**	0.00	↓**-0.94**	↓**-0.18**	-0.06	-0.04
		Prefix-preserving (PP)	↓-0.37	↓ **-0.66**	+0.01	↓**-0.94**	↓**-0.18**	0.00	**-0.01**
	DP-Noise	$\epsilon = 0.1$	↓-0.06	↓-0.06	↓**-0.11**	+0.02	+0.01	0.00	-0.08
		$\epsilon = 1$	↓ -0.06	↓ -0.14	↓-0.08	↓-0.01	0.00	↑**+0.02**	-0.08
Single-flow	No-mitigation		0.37	0.99	0.12	0.94	0.35	0.18	0.79
	Anonymization	Complete anonymization (CA)	↓**-0.37**	↓**-0.99**	0.00	↓**-0.94**	↓**-0.35**	-0.05	**-0.02**
		Pseudonymization (PS)	↓-0.36	↓**-0.99**	0.00	↓**-0.94**	↓**-0.35**	-0.03	**-0.02**
		Prefix-preserving (PP)	↓-0.24	↓**-0.99**	+0.02	↓**-0.94**	↓**-0.35**	0.00	-0.04
	DP-Noise	$\epsilon = 0.1$	↓-0.08	↓-0.01	↓**-0.10**	0.00	0.00	0.00	-0.07
		$\epsilon = 1$	0.01	↓-0.02	↓-0.02	↓-0.06	↓-0.01	↑**+0.01**	-0.07

Table 6: Impact of mitigation techniques on privacy and utility averaged across IoT and SR datasets. For each column, bold values indicate the most ideal case (either biggest drop in privacy risks or lowest drop in data utility). Highlighted values denote the original results before mitigation. Non-highlighted show the absolute change relative to the original results. For EMD and Fidelity, changes are reported in negated form for consistency.

identification, with changes remaining near zero across these attacks. However, DP noise was not applied to IP or MAC address fields; therefore, these extraction and identification results should not be interpreted as evidence that DP protects these identifiers. Although MIA success decreases slightly, the reduction is substantially smaller than that achieved by anonymization-based methods. However, DP-noise is more effective at reducing privacy risks of sensitive network properties. DP-noise reduces EMD for both multi-flow and single-flow setting with lower privacy budget (stronger privacy) generally produce larger EMD reductions.

With respect to utility, DP-noise generally has little impact on fidelity but causes a larger reduction in downstream accuracy compared to anonymization. One possible explanation is that noise addition perturbs non-identifier features that are important for downstream tasks, thereby affecting task accuracy more directly.

8 Discussion

Limitations Our extraction analysis identifies contiguous matches of at least 10 tokens in each generator's native representation, rather than comparing parsed packet fields. Because token granularity differs across models, a 10-token match does not represent the same amount of information; thus, extraction rates indicate within-model memorization and are not directly comparable across architectures. Additionally, our topology analysis does not consider cross-flow correlation attack which could amplify topology leakage by exploiting relationships across flows.

Privacy From Multiple Perspectives Our findings highlight the need to evaluate privacy beyond MIA and data extraction. Quantifying real-world privacy risks in network traffic is essential for designing effective mitigations. For example, DP noise alone does not prevent data extraction or identifier memorization, while anonymization alone does not prevent inference of sensitive distributions.

The Dual Nature of Privacy and Fidelity. Beyond privacy, synthetic traffic must retain sufficient utility for downstream applications. However, privacy and fidelity are often in tension: features that improve fidelity can also increase privacy risks. For example, accurately reproducing flow durations, packet inter-arrival times, or port usage may benefit intrusion detection and QoS analysis, but can also enable adversaries to infer sensitive network topology and user behavior. Thus, improving fidelity may increase privacy risks, while stronger protections may reduce utility, requiring careful consideration of the use case and threat model.

Context-Aware Privacy-Utility Tradeoff. Ultimately, the choice of traffic generators, evaluation metrics, and mitigation techniques should depend on the specific use case, threat model, and sharing context. For internal use in trusted environments, fine-grained models such as NetSSM can preserve rich packet-level features to support comprehensive analysis. In contrast, external or public sharing requires stronger protections, including coarser-grained generators such as NetDiffusion and NetShare, anonymization of critical infrastructure, and differential privacy noise on sensitive features, albeit with reduced utility.

9 Conclusion

This paper systematically evaluates privacy risks in synthetic network traffic using membership inference, data extraction, and network-specific attacks. Across four generators and five datasets, we show that synthetic traces can leak membership, memorize training data, and reproduce network identifiers and communication structure. We further examine how attacker access, training duration, data diversity, and mitigations shape these risks and the privacy–utility tradeoff. Overall, our findings show that synthetic network traces should be treated as security-sensitive data, not inherently safe substitutes for raw traffic.

Code Availability

We release our code and evaluation framework to support reproducibility and facilitate future research. The code is available at https://github.com/VanTranHong/IMC-26-privacy-traffic-generator.

Acknowledgement

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A Ethics

This work does not raise any ethical issues.

B Content generated by AI

Various tools, including Copilot, Claude, ChatGPT, and Gemini, supported the preparation of this manuscript through editing, grammar checks, and retrieval of background information. All generated text was reviewed and revised by the authors, and the core ideas, analyses, and conclusions remain entirely the authors' own. The same tools were also used during code development and early-stage manuscript reviews.

C Additional topology leakage experiment

Traffic Dataset	Node-overlap	Edge overlap
IoT	33/372 (8.87%)	29/386 (7.51%)
SR	21/70 (30.00%)	18/73 (24.66%)
VNAT	8/26 (30.77%)	4/19 (21.05%)
USTC	10/1349 (0.74%)	9/1364 (0.66%)
CIC	24/1307 (1.84%)	27/1315 (2.05%)

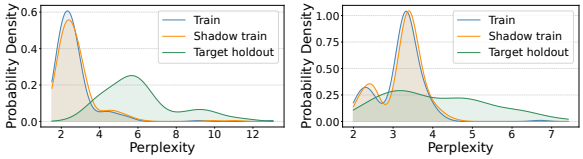
Table 7: Node and edge overlap rates between real and synthetic topologies when topology is constructed from both large and small flows. Results are reported as overlap count / total count (%).

Setting: To evaluate topology leakage across flows of varying sizes, we first identify the top 10 source IP addresses ranked by total packet volume and construct topology graphs using all flows (both large and small) originating from these sources. After generating synthetic traffic (NetSSM is trained for 1 epoch only), we extract flows associated with the same source IPs from the synthetic output and construct corresponding topologies.

Results: Compared to the setting in Section 3.1.3, where topologies are constructed only from large flows, constructing topologies from both large and small flows results in slightly lower node and edge overlap rates. This reduction is more noticeable for datasets such as IoT, USTC, and CIC, which contain a large number of nodes and edges. Nevertheless, the overlap remains substantial for datasets such as SR and VNAT, which have smaller and simpler topologies that are easier for the model to memorize. Detailed results are shown in Table 7.

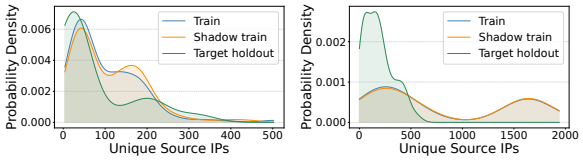
We further analyze which edges are more likely to be reconstructed in the generated outputs. Our results show that flows with higher packet counts are more likely to be memorized. Figure 8 illustrates the distribution of memorized and non-memorized edges for the SR dataset. In particular, memorized edges typically correspond to flows containing at least 1000 packets in the training data.

D Additional evidence



(a) IoT dataset (b) USTC dataset

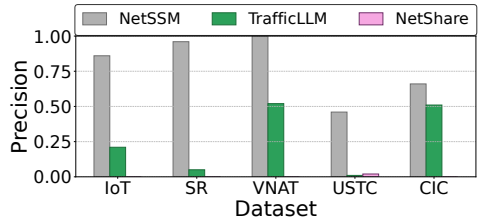
Figure 12: Probability density function (PDF) of the perplexity scores produced by NetSSM on tokens generated from the first packet of each PCAP. For the IoT dataset, the target holdout distribution is more distinguishable from the training and shadow training distributions, whereas in the USTC dataset, the target holdout distribution exhibits greater overlap with the training and shadow training distributions. Consequently, the MIA attack against NetSSM is more successful on the IoT dataset than on the USTC dataset.



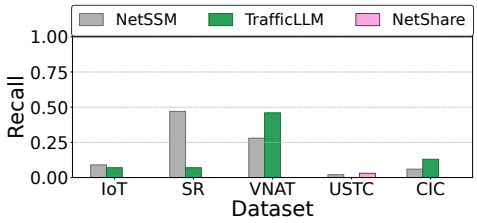
(a) IoT dataset (b) USTC dataset

Figure 13: Probability density function (PDF) of the number of unique source IPs generated by NetShare for each PCAP. For the IoT dataset, the target holdout distribution is highly similar to the training and shadow training distributions, whereas the distributions are more distinguishable in the USTC dataset. Consequently, the MIA attack against NetShare is less successful on the IoT dataset than on the USTC dataset.

E Additional results

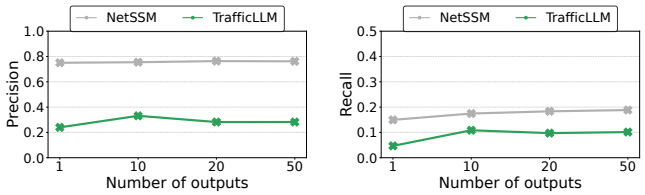


(a) Precision



(b) Recall

Figure 14: Precision and recall scores for destination IP addresses across different models and datasets. Overall, NetSSM has the highest precision and recall scores across datasets, followed by TrafficLLM. NetShare does not leak destination IP addresses.



(a) Precision (b) Recall

Figure 15: Impact of the number of generated outputs on precision and recall scores for source IP address.